

Computer Networks — Exam Solutions

Based on *Computer Networks* by Andrew S. Tanenbaum | Exam-Ready 6-Mark Format

Question 1: Discussing the limitation of "Go-Back-N", explain the advantage of "Selective Repeat" Sliding Window Protocol. [6 Marks]

1. Introduction

Both **Go-Back-N (GBN)** and **Selective Repeat (SR)** are sliding window protocols operating at the Data Link Layer and Transport Layer to ensure reliable, in-order delivery of data frames over unreliable channels.

2. Limitations of Go-Back-N (GBN) Protocol

- **Receiver Window Size is 1 ($W_r = 1$):** The receiver accepts frames *only* in strict sequential order. If frame k is lost or corrupted, all subsequent arriving frames ($k+1, k+2, \dots$) are immediately discarded, even if received correctly.
- **Massive Retransmission Overhead:** When a timeout occurs for frame k , the sender must retransmit frame k **plus all subsequent unacknowledged frames** currently in the pipeline.
- **Poor Efficiency on High BDP & Noisy Links:** On links with high propagation delay or frequent bit errors, retransmitting the entire window drastically degrades overall throughput.

GBN Retransmission Example:

```
Sender:  [Frame 0] [Frame 1] [Frame 2 (LOST)] [Frame 3] [Frame 4]
Receiver:  ACK 0      ACK 1      Discard!      Discard! Discard!
Timeout! -> Sender must retransmit: Frame 2, Frame 3, Frame 4...
```

3. Advantages of Selective Repeat (SR) Protocol

- **Receiver Window Size > 1 ($W_r > 1$):** The receiver maintains a buffer of size W_r . It accepts out-of-order frames that fall within its window and buffers them until missing lower-numbered frames arrive.
- **Selective Retransmission:** Only the specific frame that was lost or corrupted is retransmitted (triggered either by an individual timer per frame or a Negative Acknowledgment / NAK).
- **High Channel Utilization:** Because correctly received frames are not discarded, channel throughput remains high even over noisy or high-latency connections.

Selective Repeat Example:

```
Sender:  [Frame 0] [Frame 1] [Frame 2 (LOST)] [Frame 3] [Frame 4]
Receiver:  ACK 0      ACK 1      (Expecting 2)      Buffered Buffered
NAK 2 sent -> Sender retransmits ONLY Frame 2.
```

4. Mathematical Constraint (Tanenbaum's Rule)

To avoid frame ambiguity (where a receiver cannot distinguish between a fresh frame and a duplicate), the window size W in Selective Repeat must satisfy:

$W_s + W_r \leq 2^m$ & implies; $W \leq 2^{m-1}$ (where m is the number of sequence number bits)

Question 2: Compare Virtual-Circuit and Datagram Networks on the basis of important performance issues. [4/6 Marks]

1. Conceptual Overview

Datagram Subnets: Connectionless approach where each packet is treated independently and routed on an ad-hoc basis (e.g., Internet Protocol / IP).

Virtual-Circuit Subnets: Connection-oriented approach where a logical path is pre-established before any data transfer begins (e.g., MPLS, ATM, Frame Relay).

2. Comprehensive Comparison Table

Performance Issue	Datagram Subnet	Virtual-Circuit (VC) Subnet
Circuit Setup	No setup delay. Packets are sent immediately without prior signaling.	Setup delay required. Traffic cannot flow until the connection setup phase completes.
Addressing Overhead	High. Every packet must carry full source and destination IP addresses.	Low. Packets carry only short local Virtual Circuit Identifiers (VCIDs).
State Information	Stateless. Routers keep no state information about individual connections.	Stateful. Routers maintain connection tables for all active virtual circuits.
Routing / Path	Dynamic. Each packet is routed independently; different packets may take different routes.	Fixed. Path is decided during setup; all packets follow the exact same route.
Router Failures	Highly Resilient. Packets automatically bypass failed routers via dynamic recalculation.	Vulnerable. If a router along the circuit fails, all VCs through it are terminated.
Congestion Control	Difficult. Traffic flows unpredictably across multiple paths, making resource reservation hard.	Easy. Resources (bandwidth, buffer space) can be reserved in advance during setup.
Quality of Service (QoS)	Best-Effort. Hard to guarantee latency or throughput bounds per connection.	Guaranteed. Well-suited for real-time applications (voice/video) via resource reservation.

Question 3: Compare OSI and TCP/IP Network Architecture models in terms of Protocol Implementation issues. [4/6 Marks]

1. Separation of Services, Interfaces, and Protocols

- **OSI Model:** Maintains a strict, explicit distinction between *Services* (what a layer does), *Interfaces* (how upper layers access it), and *Protocols* (how the layer works internally). This makes replacing protocols theoretically simpler.
- **TCP/IP Model:** Did not clearly separate these concepts. The architecture was designed *after* the protocols were already implemented, leading to tighter coupling and less flexibility for updates.

2. Connection-Oriented vs. Connectionless Support

- **OSI Model:** Supports **both** Connectionless and Connection-Oriented in the Network Layer, but **only Connection-Oriented** in the Transport Layer.
- **TCP/IP Model:** Supports **only Connectionless** in the Network Layer (IP), but **both modes** (TCP and UDP) in the Transport Layer.

3. Layer Redundancy & Overhead

- **OSI Model:** Defines 7 layers (including Session and Presentation). In practical software implementations, Session and Presentation features are rarely implemented as distinct layers, creating unnecessary processing overhead and redundant error control.
- **TCP/IP Model:** Combines Application, Presentation, and Session into a single flexible **Application Layer** (4 layers total), optimizing performance and simplifying protocol stack implementation in operating systems.

4. The "Bad Timing" & "Bad Technology" Critique (Tanenbaum)

OSI protocols were standardized too late. By the time OSI standards were finalized, TCP/IP was already heavily deployed in universities and industry (ARPANET). OSI protocols (e.g., TP4) were overly complex and performed poorly compared to TCP/IP stacks.

Question 4: Data Frame Transmission with CRC Error Detection [6 Marks]

1. Input Parameters

- **Original Data Frame (D):** 111000111000 (12 bits)
- **Generator Polynomial G(x):** $X^5 + X^4 + X^2 + X + 1$

2. Generator Binary Representation

Writing G(x) in coefficient form: $1 \cdot X^5 + 1 \cdot X^4 + 0 \cdot X^3 + 1 \cdot X^2 + 1 \cdot X^1 + 1 \cdot X^0$

G(x) in Binary = 110111 (Degree $r = 5$ check bits).

3. Appending Zero Bits

Append $r = 5$ zero bits to data frame D: **1110001110000000**

4. Modulo-2 Division (Bitwise XOR)

```

          101110011101 <-- Quotient
110111 | 111000111000000000
        110111
        -----
          011111
          000000
          -----
            111111
            110111
            -----
              010001
              000000
              -----
                100011
                110111
                -----
                  101000
                  110111
                  -----
                    111110
                    110111
                    -----
                      010010
                      000000
                      -----
                        100100
                        110111
                        -----
                          100110
                          110111
                          -----
                            100010
                            110111
                            -----
                              101010
                              110111
                              -----
                                11101 <-- Remainder (CRC)
```

- **Calculated CRC Remainder: 11101**

- **Transmitted Frame T(x):** $111000111000 + 11101 = 11100011100011101$

5. Conditions for Acceptance vs. Retransmission

1. **Condition for Acceptance:** Upon receiving frame $T'(x)$, the destination node divides $T'(x)$ by $G(x) = 110111$ using modulo-2 division. If the remainder is **all zeros (00000)**, the frame is assumed error-free and is **ACCEPTED**.
2. **Condition for Retransmission:** If the remainder is **non-zero ($\neq 00000$)**, an error occurred during transmission. The destination node **DISCARDS** the frame and requests retransmission (via NAK or timeout).